```
import pandas as pd
```

```
df=pd.read_csv('/content/titanic.csv')
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alo
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	Fa
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	T
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	T
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	T
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	T
888	0	3	female	NaN	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	Fa
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	T
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	T

891 rows × 15 columns

Next steps: Generate code with df | New interactive sheet

```
# if we want only top data then we use head()
```

```
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Next steps: Generate code with df | New interactive sheet

```
#if we want last data then we will use tail()
```

```
df.tail()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
886	0	2	male	27.0	0	0	13.00	S	Second	man	True	NaN	Southampton	no	True
887	1	1	female	19.0	0	0	30.00	S	First	woman	False	B	Southampton	yes	True
888	0	3	female	NaN	1	2	23.45	S	Third	woman	False	NaN	Southampton	no	False
889	1	1	male	26.0	0	0	30.00	C	First	man	True	C	Cherbourg	yes	True
890	0	3	male	32.0	0	0	7.75	Q	Third	man	True	NaN	Queenstown	no	True

```
df.shape
```

```
(891, 15)
```

```
df.sample(3)  #it will return random indexed data
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
494	0	3	male	21.0	0	0	8.05	S	Third	man	True	NaN	Southampton	no	True
282	0	3	male	16.0	0	0	9.50	S	Third	man	True	NaN	Southampton	no	True
143	0	3	male	19.0	0	0	6.75	Q	Third	man	True	NaN	Queenstown	no	True

```
# if we want to check total col in a dataframe
```

```
df.columns
```

```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='object')
```

```
#if we want to check data type of col
df.dtypes
```

	0
survived	int64
pclass	int64
sex	object
age	float64
sibsp	int64
parch	int64
fare	float64
embarked	object
class	object
who	object
adult_male	bool
deck	object
embark_town	object
alive	object
alone	bool

dtype: object

```
#if we want to check overall info then we can use df.info()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    object
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    object
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), float64(2), int64(4), object(7)
memory usage: 92.4+ KB
```

```
df.head(2)
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False

Next steps: Generate code with df | New interactive sheet

```
#return a nwe dataframe containing only categorical and object column
categorical_df=df.select_dtypes(include=['category','object'])
categorical_df.columns.tolist()
```

```
['sex', 'embarked', 'class', 'who', 'deck', 'embark_town', 'alive']
```

```
#option 1
#return a new dataframe containing only int column
int_df=df.select_dtypes(include=['category','int'])
int_df.columns.tolist()
```

```
['survived', 'pclass', 'sibsp', 'parch']
```

```
#option 2
#extract all int col
df_int=df.select_dtypes(include='int')
df_int.columns.tolist()
```

```
['survived', 'pclass', 'sibsp', 'parch']
```

```
df.head(2)
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False

Next steps: Generate code with df | New interactive sheet

```
# if we wanr to check missing data in adf thrn we will use df.isnull().sum()
```

```
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

```
df['age']=df['age'].fillna(df['age'].mean())
```

```
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	0
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

```
df['age'].mean()
```

```
np.float64(29.69911764705882)
```

```
df.head(2)
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False

Next steps: Generate code with df | New interactive sheet

```
#if i want to check total sub-categories in a col then we will use df.value_counts()
```

```
df['sex'].value_counts()
```

sex	count
male	577
female	314

dtype: int64

```
df['class'].value_counts()
```

class	count
Third	491
First	216
Second	184

dtype: int64

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